

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, APRIL 2021**Mathematics****ME 2C 02—MATHEMATICAL ECONOMICS****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

ME 2C 02—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. _____ is a measure of inequality in variables.
(A) Binomial distribution. (B) Poisson distribution.
(C) Pareto distribution. (D) None of these.
2. While drawing Lorenz curve, cumulative percentage of income is taken along the :
(A) X-axis. (B) Y-axis.
(C) XY plane. (D) None of these.
3. The farther the Lorenz curve from the line of equal distribution, the _____ inequality in income.
(A) Lesser (B) Poorer.
(C) Greater. (D) Moderate.
4. LPP the transportation problem was contributed by :
(A) Leontief. (B) Koop man.
(C) G.B. Dantzig. (D) None.
5. Which of the following is an assumption of LP ?
(A) Certainty. (B) Divisibility.
(C) Additivity. (D) All the above.
6. LP is a quantitative technique of decision making using _____ constraints.
(A) Inequality. (B) Equality.
(C) Both (A) and (B) (D) None.
7. One of the limitations of LPP is to satisfy the assumption of _____ of objective function and constraints.
(A) Certainty. (B) Continuity.
(C) Linearity. (D) None.

8. If no feasible solution of the problem exists, then that LPP is said to be :
- (A) Unbounded. (B) Bounded.
(C) Infeasible. (D) Feasible.
9. A system has 3 equations with 3 variables. If it is solvable, such a solution is called :
- (A) Optimal solution. (B) Basic solution.
(C) Multiple solution. (D) None.
10. A system with m equations and n variables has at most _____ basic solutions.
- (A) $n-m$. (B) $m-n$.
(C) cCm . (D) None of these.
11. If the primal problem is a maximisation problem, then the dual problem is a _____ problem.
- (A) Minimisation. (B) Transportation.
(C) Assignment. (D) None.
12. The method of solving LPP with greater than or equal to type constraints is :
- (A) Two phase method. (B) M method.
(C) Both (A) and (B). (D) None.
13. Game theory is really the _____.
- (A) Conflict. (B) Science of conflict.
(C) Competition. (D) None.
14. When there is no saddle point in a game, it will be the case of ?
- (A) Pure strategy. (B) No strategy.
(C) Mixed strategy. (D) None of these.
15. An input-output model in which some of the production is consumed by external bodies is called :
- (A) Open. (B) Closed.
(C) Both (A) and (B). (D) None.

16. When the Lorenz Curve coincides with the line of perfect equality, it indicates ?
- (A) No inequality.
 - (B) 100% inequality.
 - (C) 50% inequality.
 - (D) None of the above.
17. For a LPP, if there is a tie in the net evaluation row of the simplex table corresponding to an iteration :
- (A) The tie should be broken before proceeding further.
 - (B) We can choose any one and proceed.
 - (C) It means something has gone wrong.
 - (D) The solution may cycle round.
18. In linear programming the number of constraints in the dual of a given primal problem is :
- (A) The number of variables in the primal.
 - (B) The number of constraints in the primal.
 - (C) The number of constants in the primal.
 - (D) None of the above.
19. A constant sum game is one in which :
- (A) The total gain is constantly changing.
 - (B) The total gain is indeterminate.
 - (C) The total gain is fixed.
 - (D) None of the above.
20. The pay off matrix in a game refers to :
- (A) The total pay-off in a game.
 - (B) The pay-off to one player in a game.
 - (C) The payments made by players in a co-operative game.
 - (D) None of the above.

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, APRIL 2021

Mathematics

ME 2C 02—MATHEMATICAL ECONOMICS

Time : Three Hours

Maximum : 80 Marks

Part A*Answer all the twelve questions.**Each question carries 1 mark.*

1. Define income inequality.
2. Write any two merits of Gini coefficient.
3. What are constraints ?
4. Define feasible region.
5. Define a slack variable.
6. What is a dual problem ?
7. What is a shadow price ?
8. What is a zero sum game ?
9. What is saddle point ?
10. Define Leontieff matrix.
11. State two limitations of input-out model.
12. What is the input-output technique ?

(12 × 1 = 12 marks)

Part B*Answer any six questions in two or three sentences.**Each question carries 3 marks.*

13. What is a Lorenz Curve ? Write any two limitations of Lorenz curve.
14. Write short note on Income tax and wealth tax.

Turn over

15. Write the dual of the LPP : Maximize

$$\pi = 120x_1 + 360x_2$$

subject to

$$7x_1 + 2x_2 \leq 28$$

$$3x_1 + 9x_2 \leq 36$$

$$6x_1 + 4x_2 \leq 48$$

$$x_1, x_2 \geq 0$$

16. Discuss theory of games.
17. Explain the interrelationship between linear programming and game theory.
18. Define value of a game.
19. What is input-output co-efficients ? What conditions may be filled if they are to be stable through time ?
20. What do you understand by Leontief Inverse Matrix and what does it show ?
21. Discuss the significance of input-output table.

(6 × 3 = 18 marks)

Part C

*Answer any six questions from the following.
Each question carries 5 marks.*

22. A manufacturer makes two products x_1 and x_2 . The first requires 5 hours for processing, 3 hours for assembling, and 4 hours for packaging. The second requires 2 hours for processing, 12 hours for assembling, and 8 hours for packaging. The plant has 40 hours available for processing, 60 for assembling, and 48 for packaging. The profit margin for x_1 is Rs. 7 ; for x_2 it is Rs. 21. Express the data in equations and inequalities necessary to determine the output mix that will maximize profits.
23. Solve graphically : Maximize $\pi = 8x_1 + 6x_2$ subject to $2x_1 + 5x_2 \leq 40$

$$3x_1 + 3x_2 \leq 30$$

$$8x_1 + 4x_2 \leq 64$$

$$x_1, x_2 \geq 0$$

24. Solve graphically : Minimize $c = 6y_1 + 3y_2$

subject to $y_1 + 2y_2 \geq 14$

$$y_1 + y_2 \geq 12$$

$$3y_1 + y_2 \geq 18$$

$$y_1, y_2 \geq 0$$

25. Solve the game with pay off matrix

$$M = \begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix}$$

26. Solve the game graphically with pay off matrix

$$M = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & 2 \end{bmatrix}$$

27. Illustrate the concept of Dominance with an example.
28. Write any five limitations of input-output, analysis.
29. In a two industry economy, it is known that industry I uses 10 paise of its own product and 60 paise of commodity II to produce a rupee worth of commodity I ; industry II uses non of its own product but uses 50 paise of commodity I in producing a rupees worth of commodity II and the open sector demands are Rs. 1,000 for commodity I and Rs. 2,000 for commodity II. Find the technology matrix.
30. Given the demand $F_1 = 11$ and $F_2 = 12$. Find the solution output mix for the two industries, given the input matrix

$$A = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.1 \end{bmatrix}$$

(6 × 5 = 30 marks)

Part D

*Answer any two questions from the following.
Each question carries 10 marks.*

31. Solve using simplex method : Maximize $\pi = 56x_1 + 24x_2 + x_3$

subject to $4x_1 + 2x_2 + 3x_3 \leq 240$

$$8x_1 + 2x_2 + x_3 \leq 120$$

$$x_1, x_2, x_3 \geq 0$$

32. Solve using simplex method :

$$\text{Minimize } c = 14y_1 + 40y_2 + 18y_3$$

subject to

$$2y_1 + 5y_2 + y_3 \geq 50$$

$$y_1 + 5y_2 + 3y_3 \geq 30$$

$$y_1, y_2, y_3 \geq 0$$

Turn over

33. Determine the optimum strategies of the game for the following payoff matrix

$$M = \begin{bmatrix} 1 & 2 & -1 \\ -2 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}.$$

34. In a three sector economy, the input co-efficient matrix and final demand vector are as given below :

$$A = \begin{bmatrix} 0.3 & 0.2 & 0.3 \\ 0.1 & 0.3 & 0.4 \\ 0.2 & 0.3 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 500 \\ 700 \\ 600 \end{bmatrix}$$

Find the sectoral output X_1 , X_2 and X_3 using Cramers rule.

(2 × 10 = 20 marks)